

CUSTOMER MILESTONE BRIEF

MS #27 - Offline Data Movement and CDC Rehearsal

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.27.0
Date	2026-08-28
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #27 - Offline Data Movement and CDC Rehearsal converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Adds deterministic change ordering, resume, and data-movement rehearsal before connection to a live replication stream.

Why this matters to a customer

Adds deterministic change ordering, resume, and data-movement rehearsal before connection to a live replication stream.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a content-addressed five-event Db2-shaped AUTHFRDS change journal covering insert, update, and delete operations.
- Added deterministic PostgreSQL- and Oracle-shaped target application with normalized row, checksum, and identity reconciliation.
- Forced an interruption after event two and proved exact checkpoint resume plus idempotent duplicate replay with no lost or repeated change.
- Added a plan-bound, development-only human cutover approval that cannot authorize production.
- Injected a unilateral post-cutover divergence, detected it, and restored both targets to their exact pre-cutover state identities.
- Recorded bounded zero-event fixture RPO and three-step recovery while explicitly excluding wall-clock production RPO/RTO claims.
- Added fail-closed journal, before-image, checkpoint, mapping, approval, fault, receipt, and live- overclaim tests plus Bash and PowerShell verification entry points.

- Published the rehearsal in the unified Db2/Data capability and data control-tower projections; live Db2, customer-data, real cutover, production, and mainframe-equivalence gates remain blocked.

How it differs from earlier work

MS #27 builds on MS #26 (Multi-Workload Factory Qualification). The earlier milestone established its own bounded capability; this milestone adds offline data movement and cdc rehearsal while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use offline data movement and cdc rehearsal as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Published the rehearsal in the unified Db2/Data capability and data control-tower projections; live Db2, customer-data, real cutover, production, and mainframe-equivalence gates remain blocked.

Source of record

- CHANGELOG.md release section(s): 0.27.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity